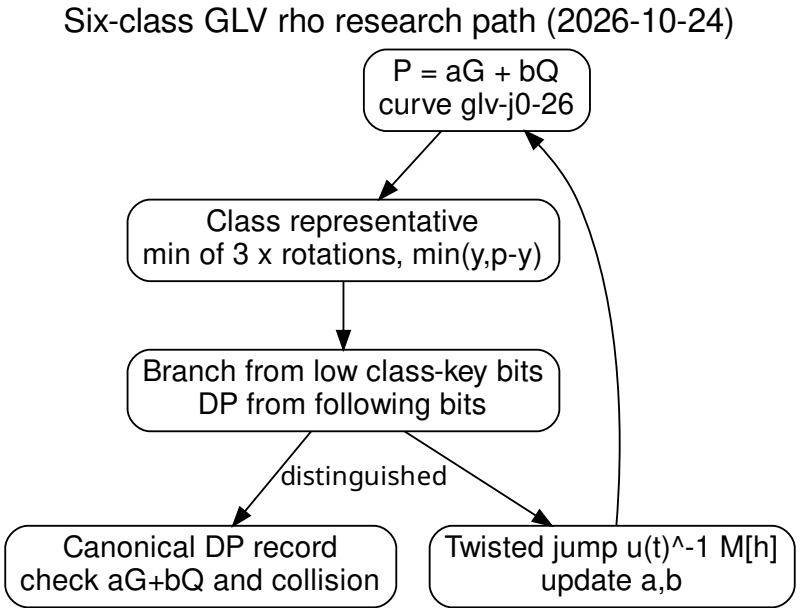
Six-class GLV rho assessment

2026-10-24

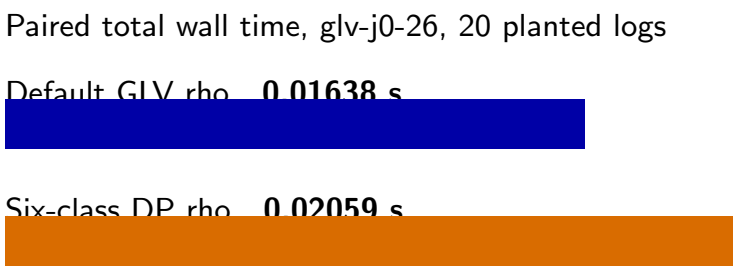
Question and method

We assessed the six-class GLV walk from Curve Smash commit 1f084c1 against our native default on registered curve `glv-j0-26`. Its class representative minimizes the three x rotations and the sign of y. The orientation twists the jump, and distinguished point records carry canonical coefficients.



Measured result

In one paired local run, both solvers recovered all 20 planted logs. Default oriented GLV rho took 0.016382582 s; six-class DP rho took 0.020590401 s, 1.257 times as long. The candidate visited 46,882 classes, emitted 2,922 DPs, and took 196 short-cycle escapes. These are total times, with no uncertainty estimate from this single run. The figure below shows that original run only.



A second local run retained default/six-class/default times per planted target. Across 20 targets, the default arms took 0.012935407 s and 0.012907277 s; six-class took 0.023595841 s, 1.826 times the default-arm mean. All answers and repeated-baseline group-operation counts matched. Each run lasted only milliseconds, so this A/A control is a diagnostic, not a confidence interval or production throughput estimate. The raw per-target rows are in `paired-glv-j0-26-aba.json`.

Interpretation and limits

This is a negative toy-curve result. It is not a secp256k1 cost measurement. Commit 1f084c1 changes κ from 1.05 to 1.02 and the inversion batch to 2^{16} while leaving the algorithm unchanged. Its 126.96-bit cost and ≥ 0.3909 success remain conditional on the stated H1 mixing and cost model. The C solver is opt-in. A Rust BigUint worker, isolated native challenge custody, a five-clause H1 matrix, a paired native comparison and one frozen follow-up protocol now provide a research path. Cairn's native checker accepts public secp256k1 distinguished points, but DP point equations do not prove walker provenance; sampled re-walks are still required. The C and Rust job derivations are different, so their receipts cannot be pooled. The H1 clauses, full-work cost and independent follow-up review remain open.

Evidence and next checks

The source record is `assessment.md`; raw receipt is `paired-glv-j0-26.json`, with A/B/A rows in `paired-glv-j0-26-aba.json`. Repeat controlled tests over the curve ladder, audit sampled trails, and build a field-operation and memory meter before assessing a fixed-limb implementation. No isogeny-route or volcano graph update follows from this walk.